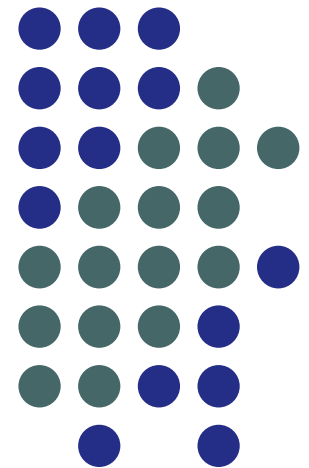


Leading Pedestrian Interval (LPI) Implementation Guidelines in Phoenix, Arizona: A Data Driven Approach

PSU Transportation Seminar



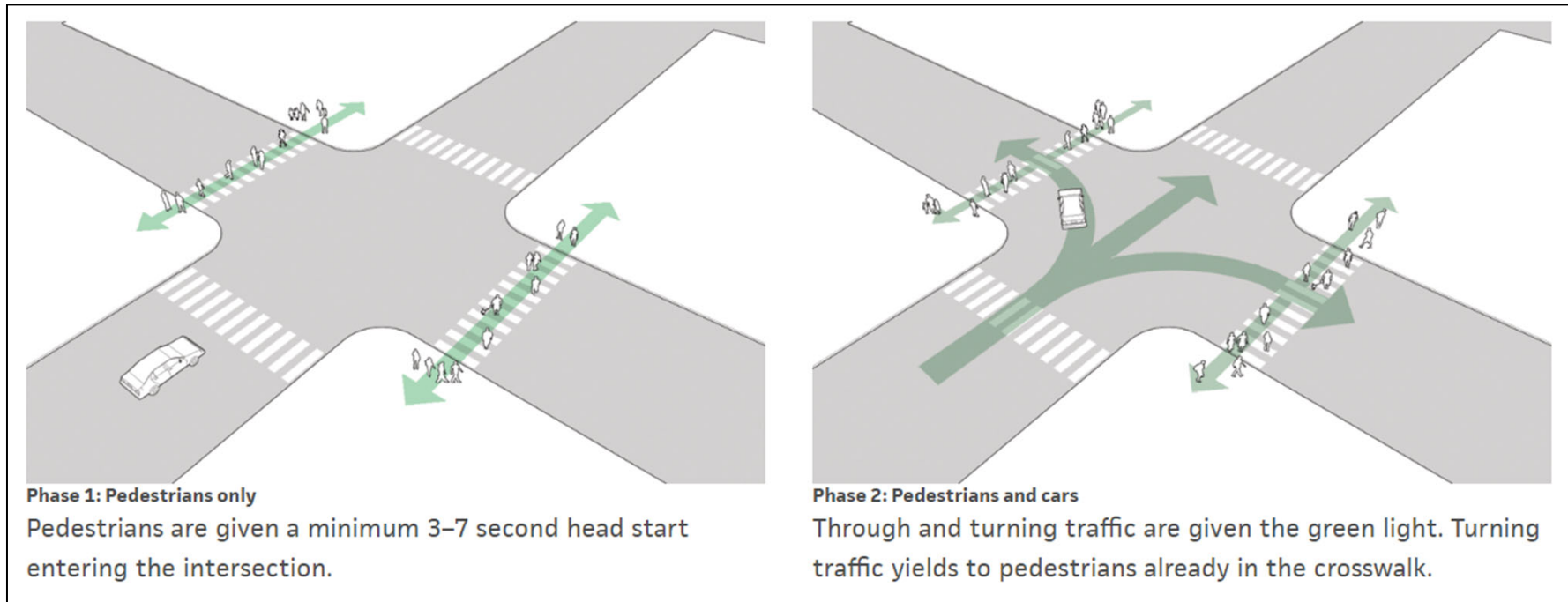
January 8th, 2026



Leading Pedestrian Interval (LPI)



- LPI provides a walk indication to a subject crosswalk prior to onset of green for the adjacent through movements.
- Reduce ped-turning vehicle conflict
- FHWA proven safety countermeasure





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<http://tampabaytrafficsafety.com/2021/09/08/fdot-leading-pedestrian-interval-lpi/>

Motivation / Process



- City of Phoenix desired LPI implementation guidance
- Process
 - Literature Review
 - Crash analysis (turning vehicle-pedestrian crashes)
 - Site identification for field data collection
 - Crash / volume analyses
 - Data collection / reduction (with / without LPI)
 - Data analysis / modeling
 - Guideline development

Crash Analysis Objectives



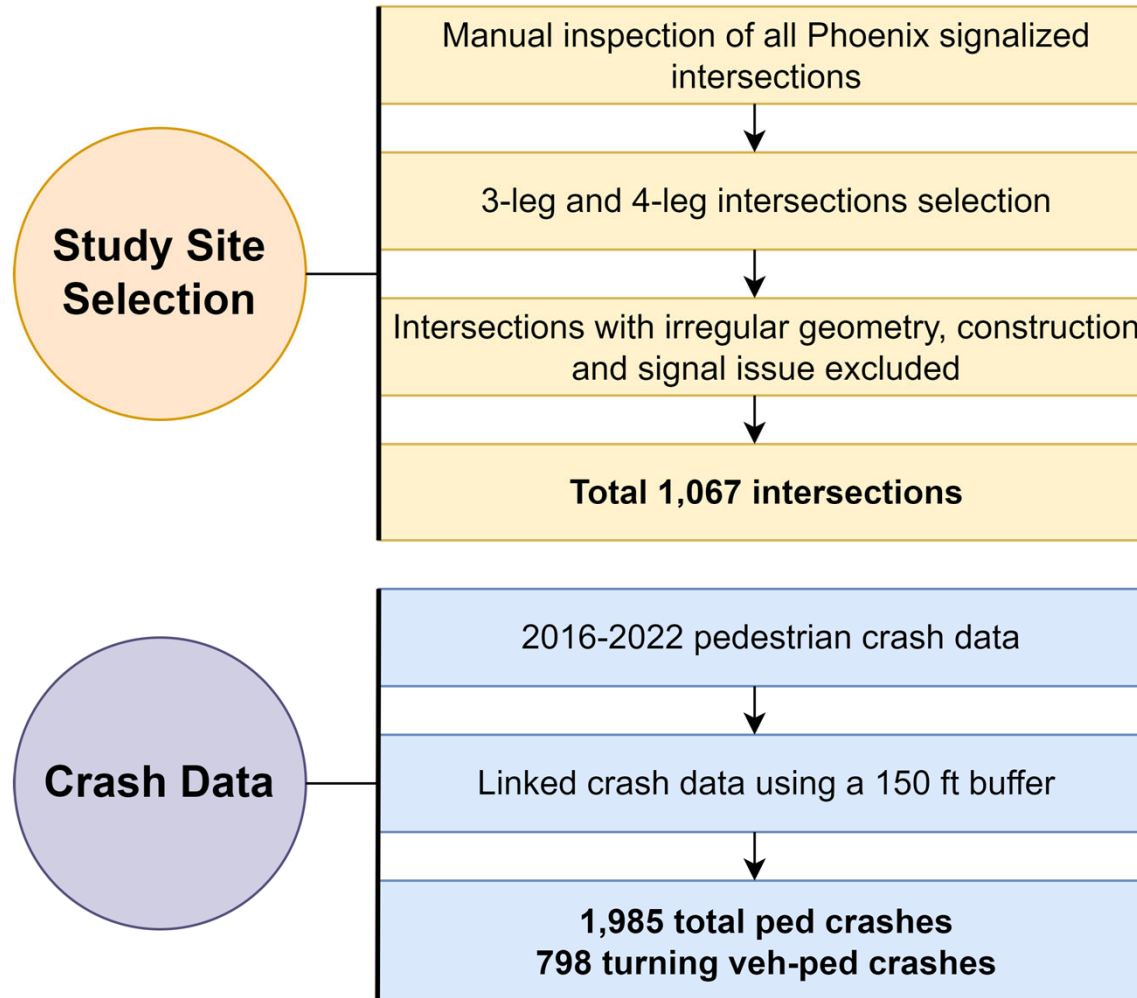
Research Context

- LPI implementation guidelines can vary from jurisdiction to jurisdiction and weighting schemes may not always be based on empirical data.
- The evaluation of these variables with empirical data and modeling can aid in creating more robust and data driven LPI guidelines.

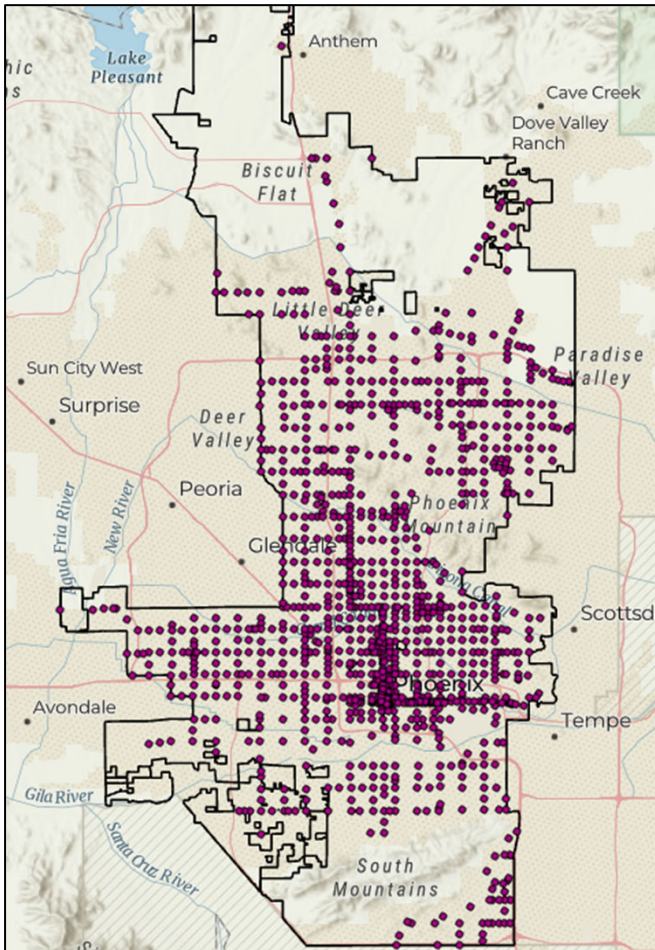
Objectives

- Develop a crash frequency model to examine the impacts of factors considered in different LPI guidelines on turning vehicle-pedestrian crashes.
- Offer insight on the factors that most impact turning vehicle-pedestrian crashes.

Data Description



Intersection Data Collection



Phoenix Signalized Intersections

Data collected from Google Maps Aerial / Street View

- T-intersection
- One-way street
- Intersection skewed
- Sight obstructions
- Median Present Through Crosswalk
- Crosswalk Length
- No Right Turn On Red (NTROR) sign
- Protected turn phase
- Exclusive turn lanes
- Push Button at intersection corners

Data collected using GIS

- Proximity to School
- Proximity to Transit Stop
- Major AADT
- Minor AADT
- Gross Activity Density

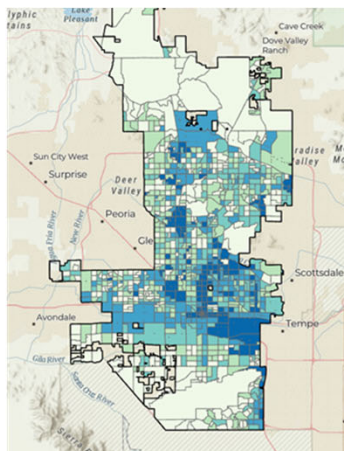
Intersection Data Collection



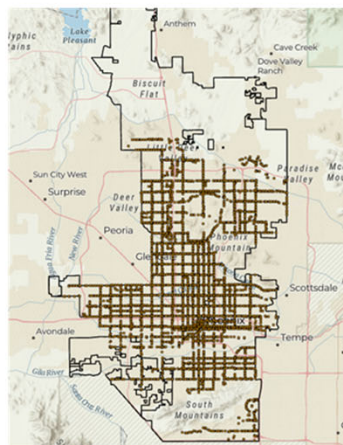
Aerial View



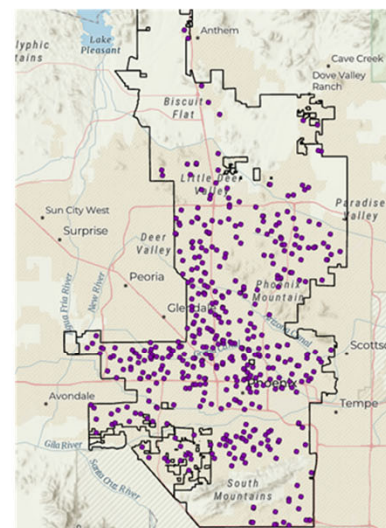
Street View



Employment Density



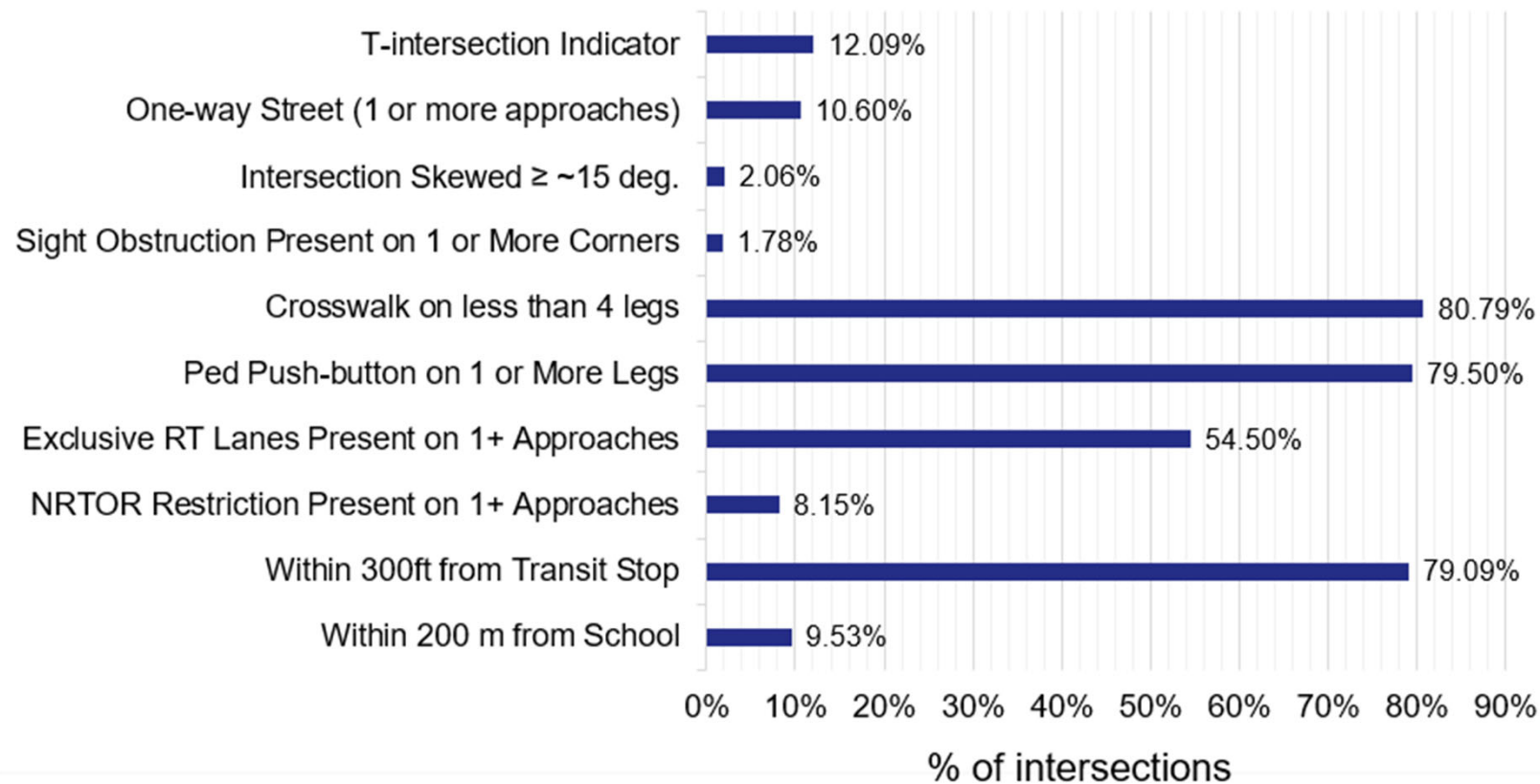
Transit Stop Locations



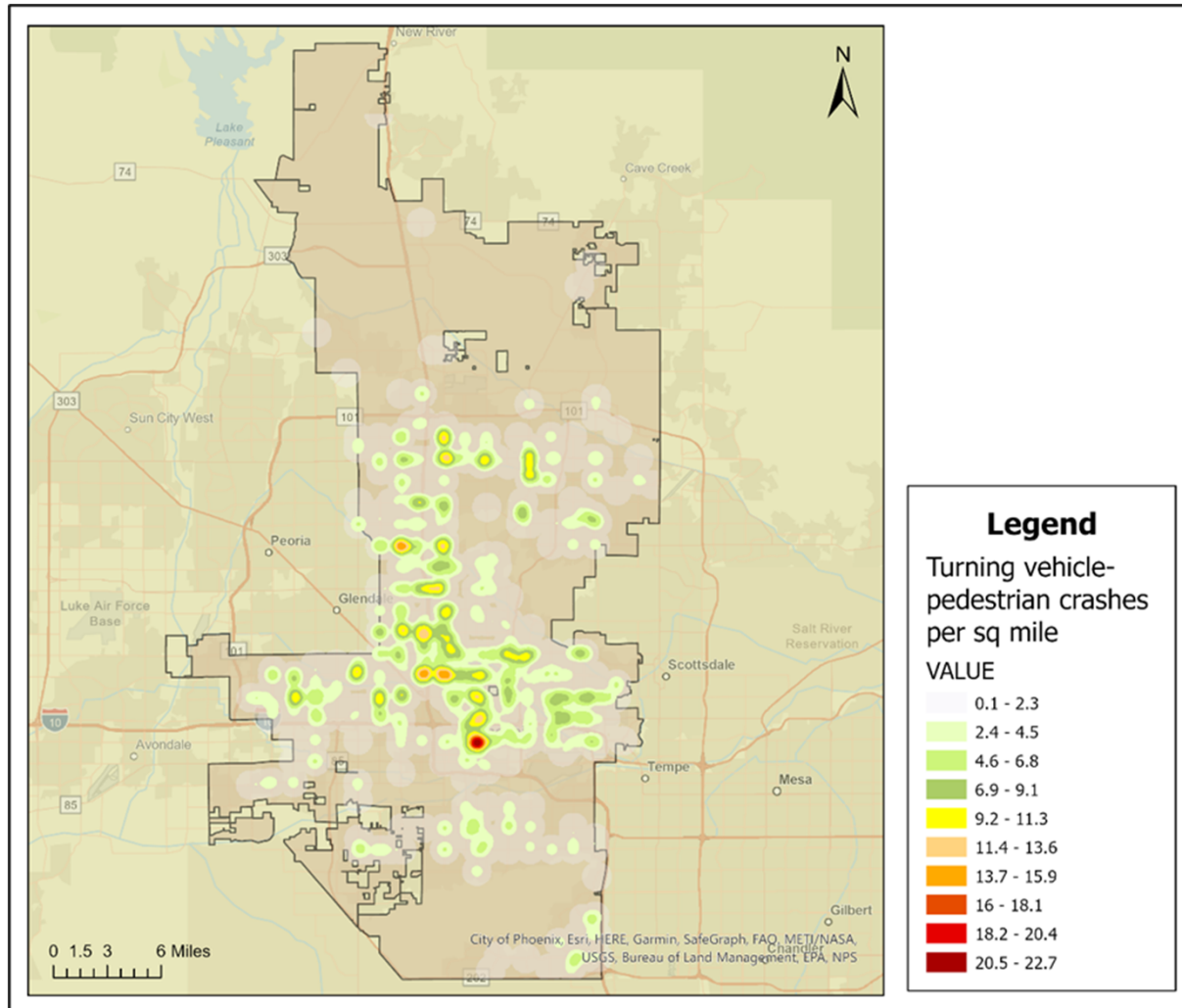
School Locations



Intersection Summary Statistics



Heat Map of Turning Vehicle-Pedestrian Crashes





Statistical Analysis Method

- Negative binomial (NB) model specification was selected for the base-level assessment of
 - (i) turning vehicle-pedestrian crashes and
 - (ii) total pedestrian crashes
- Expected percent increase in crash frequency associated with each variable was calculated.
- Model results was used to predict pedestrian crash frequency.

Model Results



Variable	Turning Vehicle-Pedestrian Crashes		Total Pedestrian Crashes	
	β	p-value	β	p-value
Constant	-4.69	<0.001***	-3.09	<0.001***
T intersection Indicator	0.42	0.167	0.38	0.091*
One way street (1 or more approaches)	-1.26	<0.001***	-0.63	<0.001***
Intersection skewed $\geq$ ~15 deg.	0.22	0.484	0.03	0.93
Site Obstruction on 1 or More Corners	0.36	0.210	-0.03	0.92
Crosswalk on 1 and 2 legs (ref.)				
Crosswalk on 3 legs	0.90	0.01***	0.52	0.04**
Crosswalk on 4 legs	1.63	<0.001***	1.29	<0.001***
Average Crosswalk Length (ft)	0.01	0.081*	0.00	0.51
No Ped Push-buttons Present (ref.)				
Ped Push-button on 1 or More Legs	-0.25	0.01***	-0.15	0.056**
RT Lanes Not Present (ref.)				
RT Lanes Present on 1+ Approaches	0.15	0.085*	0.08	0.25
NRTOR Not Present (ref.)				
NRTOR sign Present on 1+ Approaches	0.02	0.955	-0.18	0.44
Major Rd AADT 0-15,000 vpd (ref.)				
Major Rd AADT 15,001-30,000 vpd	0.18	0.274	0.25	0.055*
Major Rd AADT 30,001-45,000 vpd	0.28	0.096*	0.49	<0.001***
Major Rd AADT 45,001+ vpd	0.26	0.158	0.54	<0.001***
Minor Rd AADT 0-5,000 (ref.)				
Minor Rd AADT 5,001-15,000 vpd	0.35	0.002***	0.30	<0.001***
Minor Rd AADT 15,001-30,000 vpd	0.67	<0.001***	0.81	<0.001***
Minor Rd AADT 30,001+ vpd	0.84	<0.001***	0.86	<0.001***
Distance from School 851m+ (ref.)				
Distance from School 201-850m	0.41	<0.001***	0.37	<0.001***
Distance from School $\leq$ 200m	0.40	0.006***	0.47	<0.001***
Distance from Transit Stop 301+ ft (ref.)				
Distance from Transit Stop 0-300 ft	0.57	<0.001***	0.87	<0.001***
Gross Activity Density 0-5 per/acre (ref.)				
Gross Activity Density 5.1-15 per/acre	1.06	<0.001***	0.55	0.003***
Gross Activity Density 15.1-30 per/acre	1.50	<0.001***	0.91	<0.001***
Gross Activity Density 30+ per/acre	1.12	<0.001***	0.64	0.002***

Expected Percent Change in Crash Frequency

Variable	Turning Vehicle-Pedestrian Crashes Expected % Change
→ T intersection Indicator	51.59%
One way street (1 or more approaches)	-71.58%
Intersection skewed $\geq \sim 15$ deg.	25.22%
Site Obstruction Present on 1 or More Corners	43.36%
Crosswalk on 1 and 2 legs (ref.)	
Crosswalk on 3 legs	146.47%
→ Crosswalk on 4 legs	410.83%
Average Crosswalk Length (ft)	0.61%
No Ped Push-buttons Present (ref.)	
Ped Push-button on 1 or More Legs	-22.46%
Exclusive RT Lanes Not Present (ref.)	
Exclusive RT Lanes Present on 1+ Approaches	16.28%
NRTOR Not Present (ref.)	
NRTOR Restriction Present on 1+ Approaches	2.00%
Major Rd AADT AADT 0-15,000 vpd (ref.)	
Major Rd AADT 15,001-30,000 vpd	19.98%
Major Rd AADT 30,001-45,000 vpd	32.62%
Major Rd AADT 45,001+ vpd	29.46%
Minor Rd AADT 0-5,000 (ref.)	
Minor Rd AADT 5,001-15,000 vpd	41.55%
Minor Rd AADT 15,001-30,000 vpd	95.24%
→ Minor Rd AADT 30,001+ vpd	131.31%
Intersection Distance from School 851m+ (ref.)	
→ Intersection Distance from School 201-850m	50.63%
→ Intersection Distance from School ≤ 200 m	49.22%
Intersection Distance from Transit Stop 301+ ft (ref.)	
→ Intersection Distance from Transit Stop 0-300 ft	76.38%
Gross Activity Density 0-5 persons+jobs/acre (ref.)	
→ Gross Activity Density 5.1-15 persons+jobs/acre	189.07%
→ Gross Activity Density 15.1-30 persons+jobs/acre	346.76%
→ Gross Activity Density 30+ persons+jobs/acre	205.75%

Excluded Non-Significant Predictors

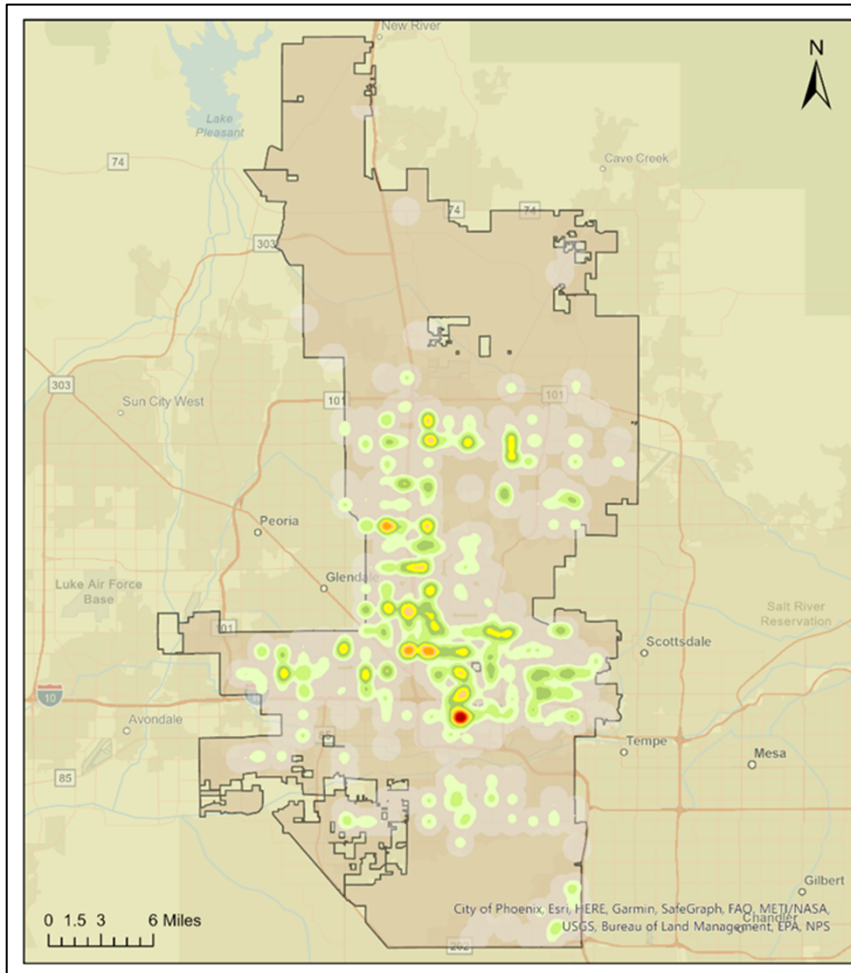


- Presence of raised medians through a crosswalk
- Presence of protected left or right turn phases
- Presence of exclusive left turn lanes
- Presence of light rail tracks through an intersection
- Presence of right turn slip lanes

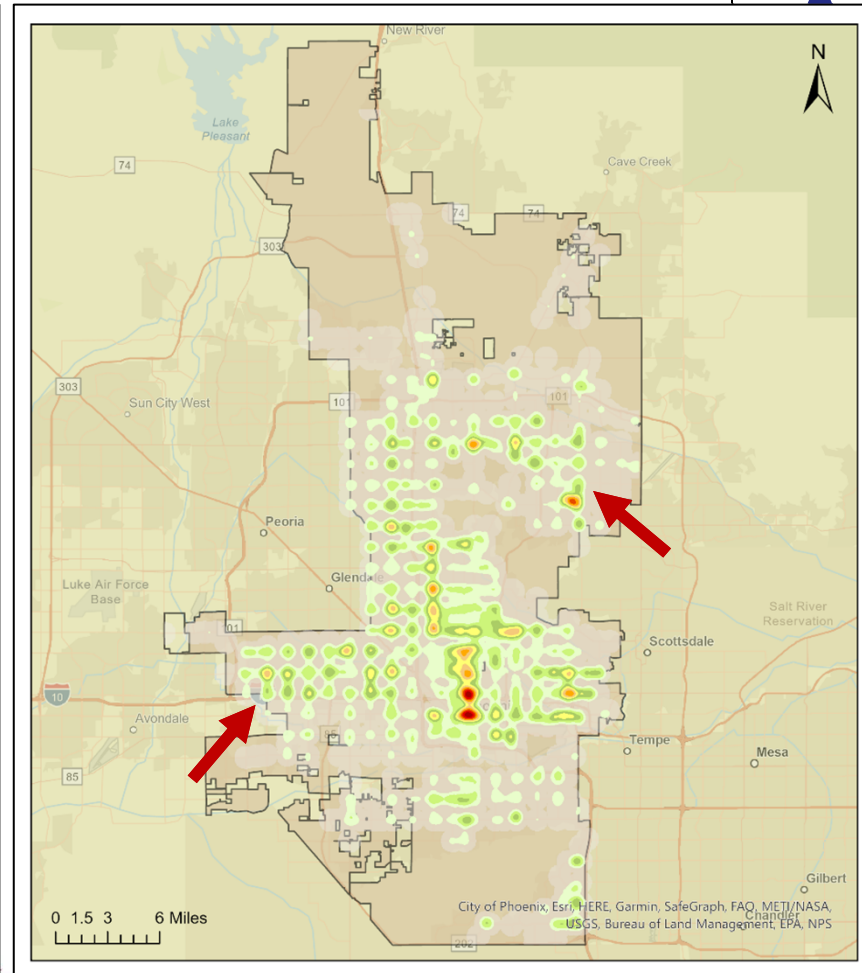
Observed vs. Predicted Crashes



Observed Crashes



Predicted Crashes



Crash Analysis Conclusions



- Significant associations found between various intersection characteristics and turning vehicle-ped crash frequency.
- The highest predicted percent increases were for the presence of four crosswalk legs, different ranges of gross activity density, higher ranges of minor road AADTs, transit stop presence, presence of a school within 850m, and the T-intersection indicator.
- Magnitude of these factors useful for developing weighting schemes for future LPI guidelines.
- Future research could explore analysis at the crosswalk level, incorporate pedestrian, and turning vehicle volume.

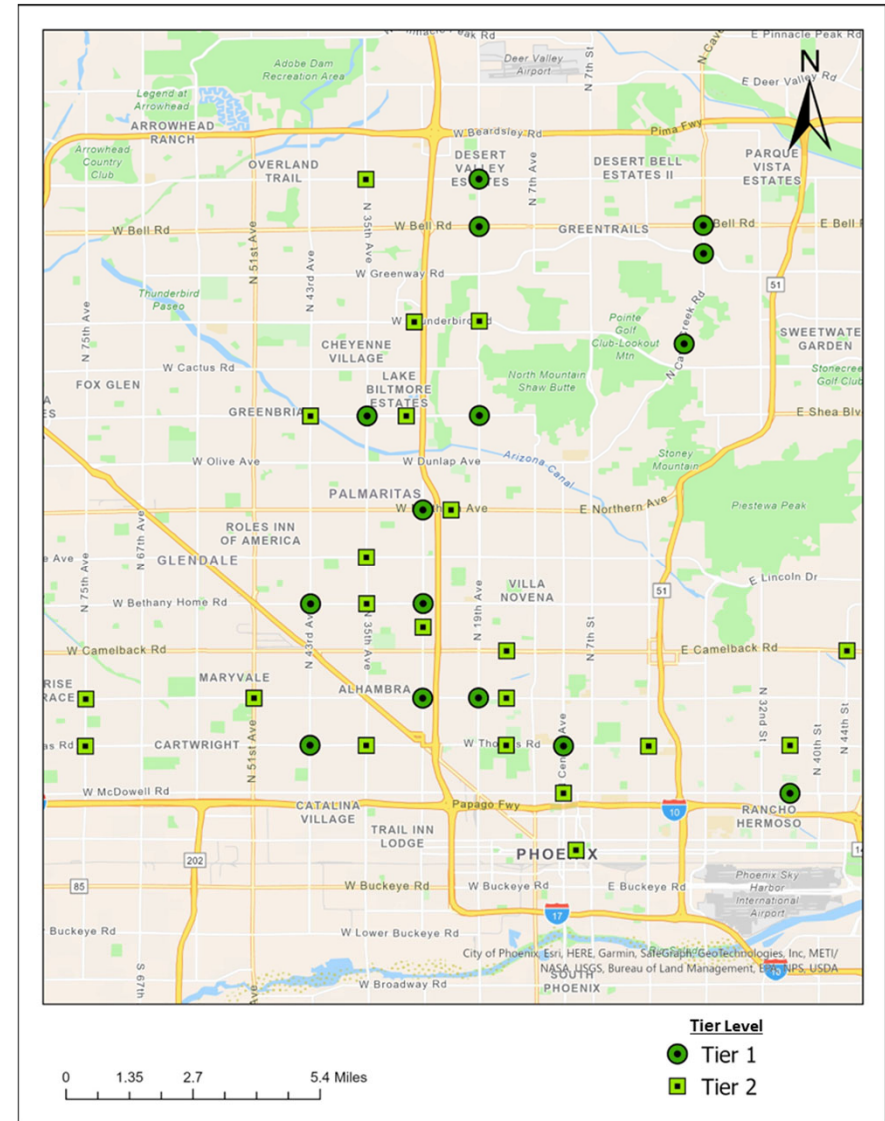
Site Identification

● Priority Intersections Identification

- Pedestrian crash data from 2016 through 2022
 - Pedestrian – turning vehicle crashes
- Ranked from highest to lowest crash frequency
- Top 85 intersections were separated into three different tiers

Selected Intersections for Volume Analysis

Tier#	Intersection	Intersection Name
1	2	Indian School & 19 th Ave
	10	Bell & 19 th Ave
	15	Camelback & 15 th Ave
	23	Indian School & 51 st Ave
2	34	Washington & 3 rd St
	60	Southern & 19 th Ave
3	65	Baseline & 51 st Ave



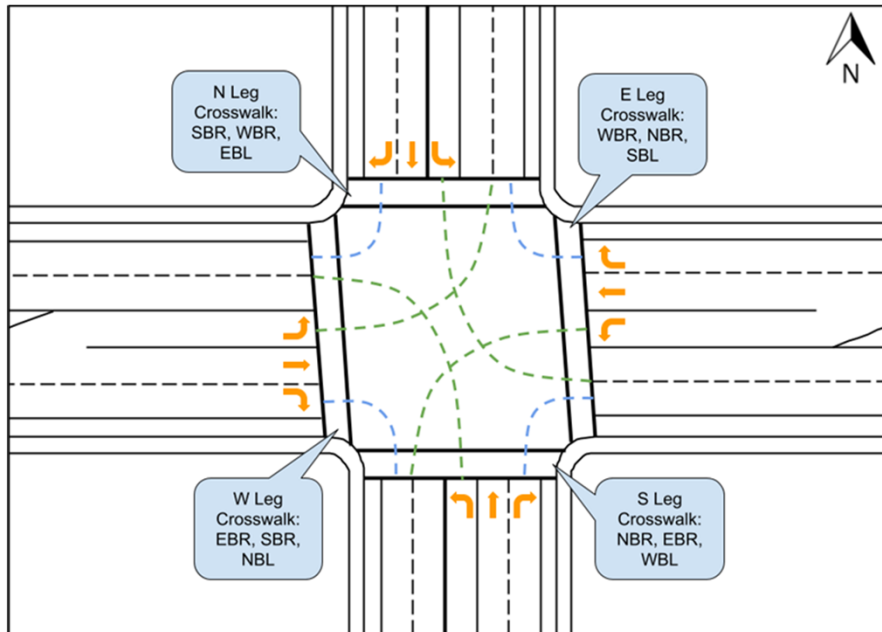
Tier 1 & Tier 2 Intersections

Site Identification

- **High Exposure Crosswalks Selection through Volume Analysis**
 - Volume Counts: 12:00 PM to 1:00 PM and 5:00 PM to 6:00 PM
 - Timing Plans to determine left turn phasing (FYA, protected only, protected-permitted, or permitted)

$$\text{CrossProduct} = \text{PedCrossing} * \text{ConflictingVehicles}$$

Conflicting Vehicles



CrossProduct of Washington/3rd

One Way Roads - Permitted Turns				
Site:		Washington & 3rd St		
Time:	Movement Leg	# Peds	# Turning Conflicting Vehicles	(#Peds) * (# Turning Conflicting Vehicles)
12:00 - 1:00 PM	N Leg	45	75	3375
	S Leg	46	182	8372
	E Leg	64	23	1472
	W Leg	100	52	5200
5:00 - 6:00 PM	N Leg	59	121	7139
	S Leg	63	229	14427
	E Leg	21	24	504
	W Leg	66	97	6402

Site:		Washington & 3rd St		
Time:	Movement Leg	# Peds	# Turning Conflicting Vehicles	(#Peds) * (# Turning Conflicting Vehicles)
12:00 - 1:00 PM	N Leg	104	196	20384
5:00 - 6:00 PM	S Leg	109	411	44799
	E Leg	85	47	3995
	W Leg	166	149	24734

Site Identification

- **Crosswalks Selection Criteria**

- High value of CrossProduct
- No more than two crosswalks from any one intersection
- Two legs at Washington and 3rd St due to the high ped volumes and agency preference

Intersection	Leg	Pedestrians Crossing	Conflicting Turning Vehicles	Cross Product
Baseline & 51st Ave	N	51	963	49113
Indian School & 19th Ave	W	52	941	48932
Washington & 3rd St	S	109	411	44799
Baseline & 51st Ave	W	38	1090	41420
Indian School & 51st Ave	S	37	1041	38517
Indian School & 19th Ave	N	42	812	34104
Indian School & 51st Ave	W	29	1099	31871
Indian School & 51st Ave	E	40	774	30960
Baseline & 51st Ave	E	29	950	27550
Indian School & 51st Ave	N	32	842	26944
Bell & 19th Ave	W	34	778	26452
Bell & 19th Ave	E	26	958	24908
Washington & 3rd St	W	166	149	24734
Indian School & 19th Ave	E	35	598	20930
Washington & 3rd St	N	104	196	20384
Indian School & 19th Ave	S	32	591	18912
Baseline & 51st Ave	S	23	812	18676
Southern & 19th Ave	E	25	699	17475
Southern & 19th Ave	S	19	802	15238
Bell & 19th Ave	S	27	386	10422
Southern & 19th Ave	W	15	617	9255
Camelback & 15th Ave	W	20	415	8300
Bell & 19th Ave	N	12	615	7380
Southern & 19th Ave	N	9	712	6408
Camelback & 15th Ave	N	18	325	5850
Camelback & 15th Ave	S	15	303	4545
Washington & 3rd St	E	85	47	3995
Camelback & 15th Ave	E	6	440	2640

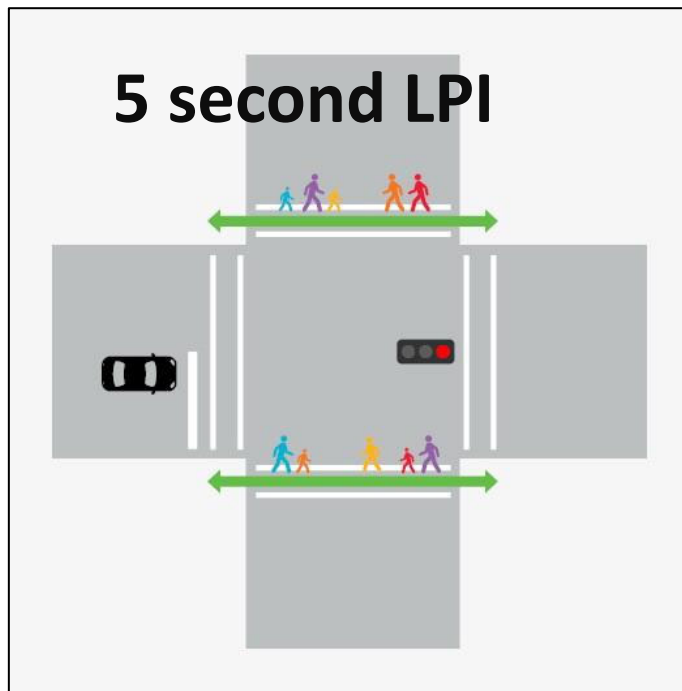
Selected Crosswalks for Data Collection





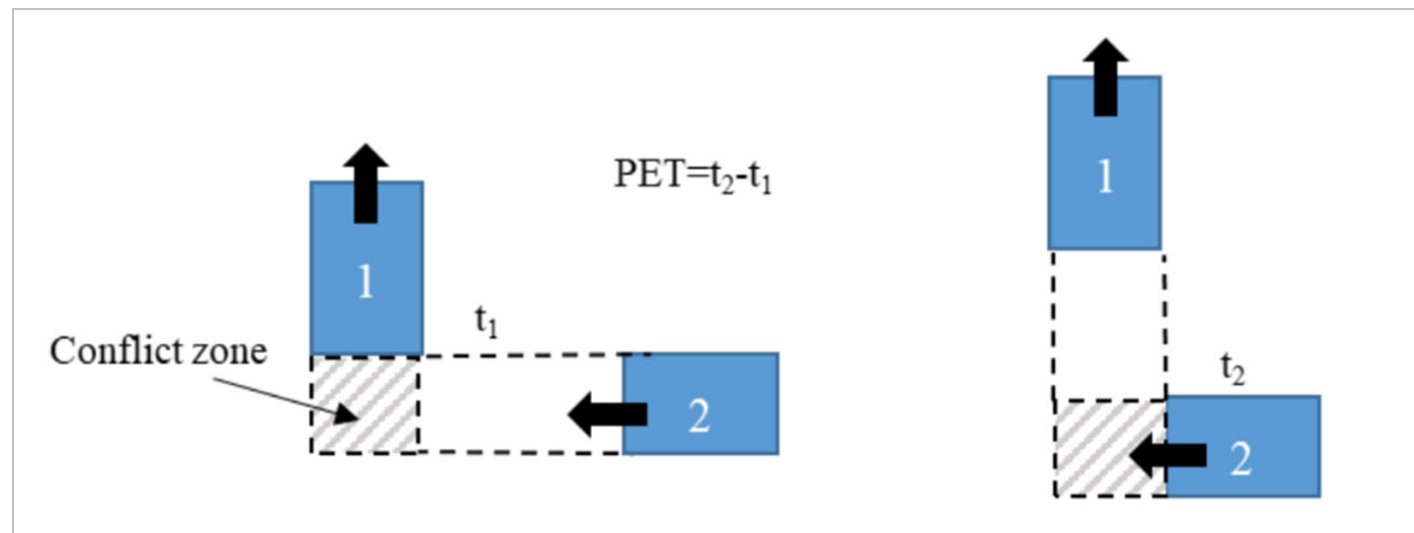
Video Data Collection

- Summertime was avoided for data collection
- **Before Phase-** Nov 8, 23 to Nov 28, 23
- **After Phase-** Feb 23, 24 to March 05, 24
- 'New Traffic Pattern Ahead' signs were installed with LPI
- Data Collection- 7AM to 5PM, 10 hours each day
- Total 160 hours of video collected



Surrogate Measure for Assessing Safety

- Indirect safety measures - assess potential risks and safety issues without relying solely on crash data
- **Post Encroachment Time (PET)** is a popular tool for assessing conflict frequency and severity
- PET measures the proximity of two road users occupying the same space at the same time, thus indicating the severity of a potential conflict.



Concept of PET

Example Interaction





Video Data Reduction

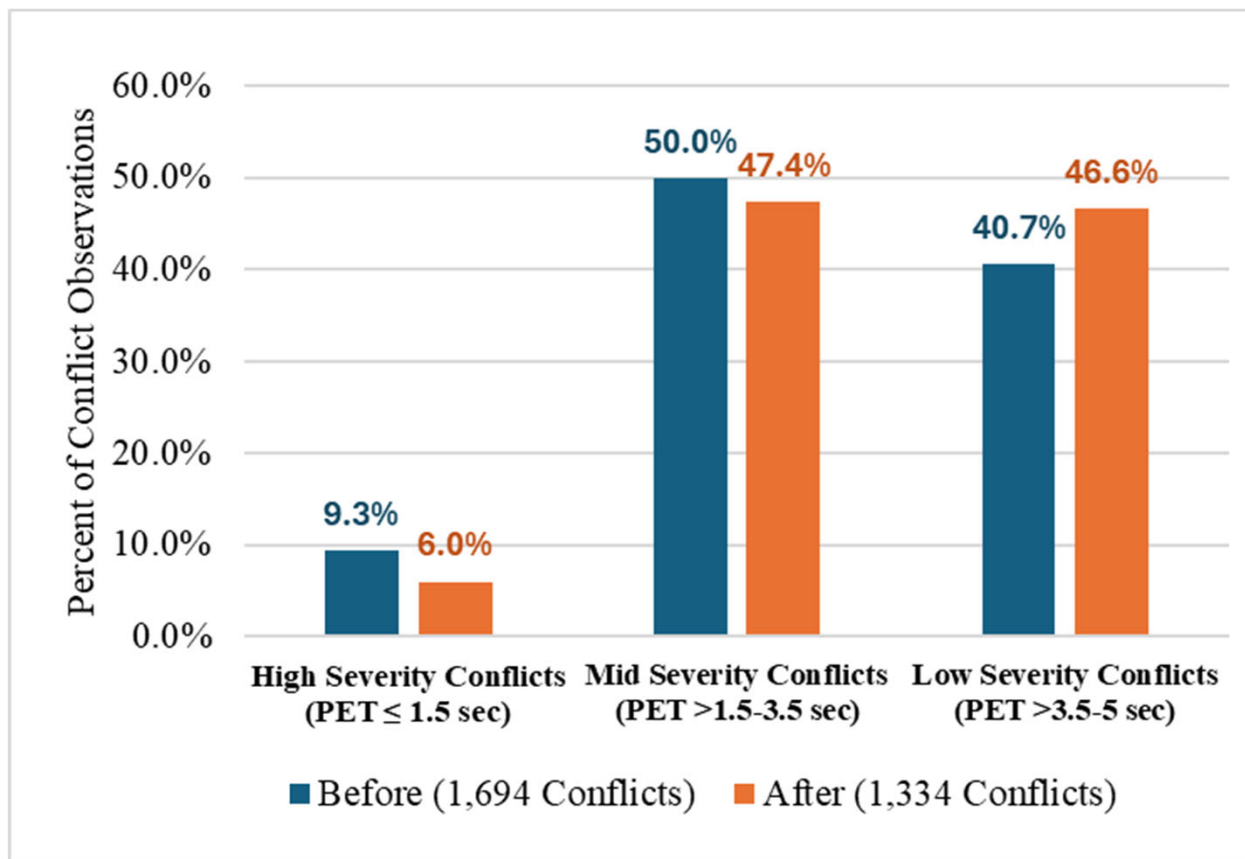
- Primarily reduced by a group of four students
- A set of instructions was provided to the team
- Trial data reduction was performed to ensure consistency



Summary of Observed Ped-Veh Conflicts



- Total Conflicts (PET $\leq$ 5 sec)
 - Before LPI – 1694 Conflicts
 - After LPI – 1334 Conflicts



Pedestrian and Vehicle Characteristics During Interactions

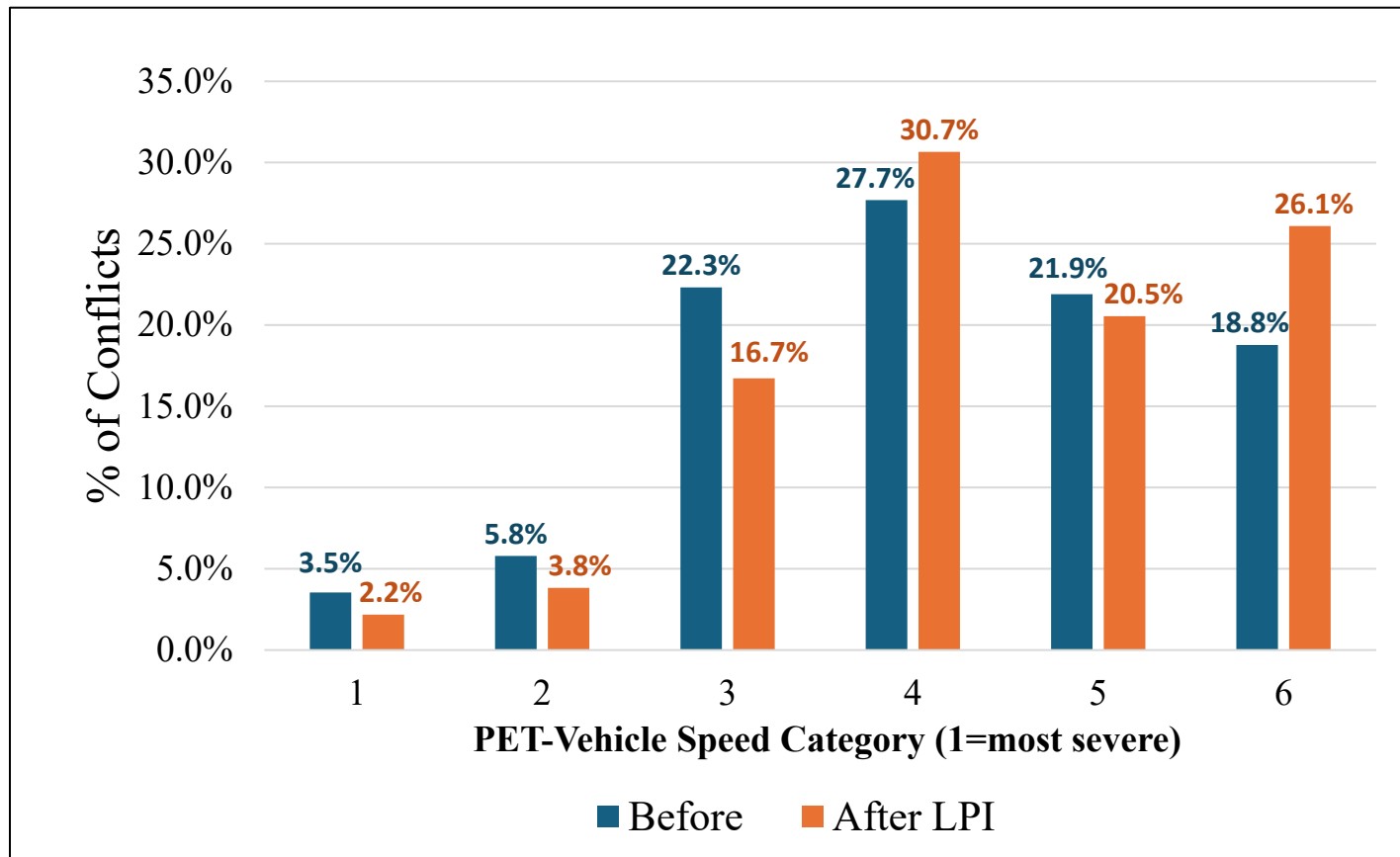


- Mean Vehicle Speed slightly reduced after LPI
- Mean Pedestrian Speed slightly increased after LPI

Conflict Characteristics		Before	After
Speed	Mean Vehicle Speed (mph)	11.61	11.35
	Mean Pedestrian Speed (ft/s)	5.43	5.61
Vehicle Direction	Conflicts with Left turning Vehicle	21.7%	21.1%
	Conflicts with Right turning Vehicle	78.2%	79.0%
First Unit	Pedestrian	68.9%	64.8%
	Vehicle	23.9%	28.4%

Conflicts by PET-Veh Speed Severity

- **Most Severe (1)** - lower PET (≤ 1.5 sec) and high vehicle speed (≥ 11.5 mph) Conflicts
- **Least Severe (6)** - higher PET (> 3.5 -5 sec) and low vehicle speed (< 11.5 mph)



Other Collected Factors

Compliance with Traffic Rules

- Driver Violations
- Driver Evasive Actions
- Pedestrian Distractions
- Pedestrian Crosswalk Violation
- Pedestrian Evasive Action

Pedestrian Demographic and Mobility Characteristics

- Pedestrian Group Size
- Pedestrian Gender
- Pedestrian Age
- Pedestrian with Additional Mobility Devices

Analysis of Pedestrian-Vehicle Conflict Frequencies

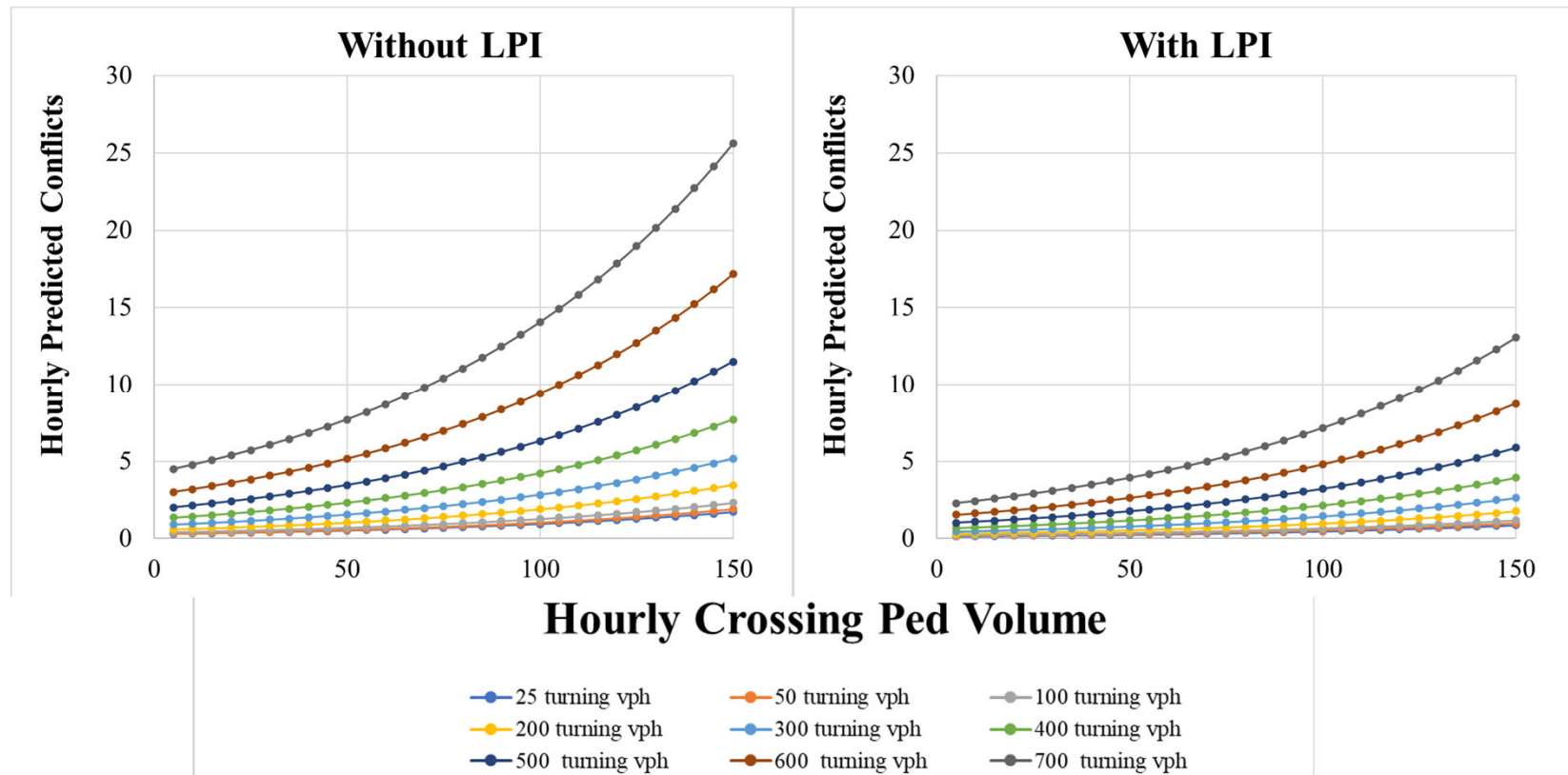


- Negative Binomial Model - non-negative conflict data
- A series of NB models were developed
 - All Reduced Conflicts ($PET \leq 5$ sec)
 - High Severity Conflicts ($PET \leq 1.5$ sec)
 - High Severity High-Speed Conflicts
- With different combinations of turning vehicle volume
 - All Turning Vehicle - Protected and Permitted Left Turns and Right Turns
 - Conflicting Turning Vehicle - Permitted Left Turns and Right Turns
 - Permitted Left Turns and Right Turns Separately

Model Predicted Counts of (PET ≤ 1.5sec) Conflicts

Predicted Conflict Equation

$$N_{\text{predicted_conflicts}} = \exp[-1.357 + (-0.673 * \text{LPI_indicator}) + (0.004 * \text{TurningVehVolume}) + (0.012 * \text{PedVolume})]$$

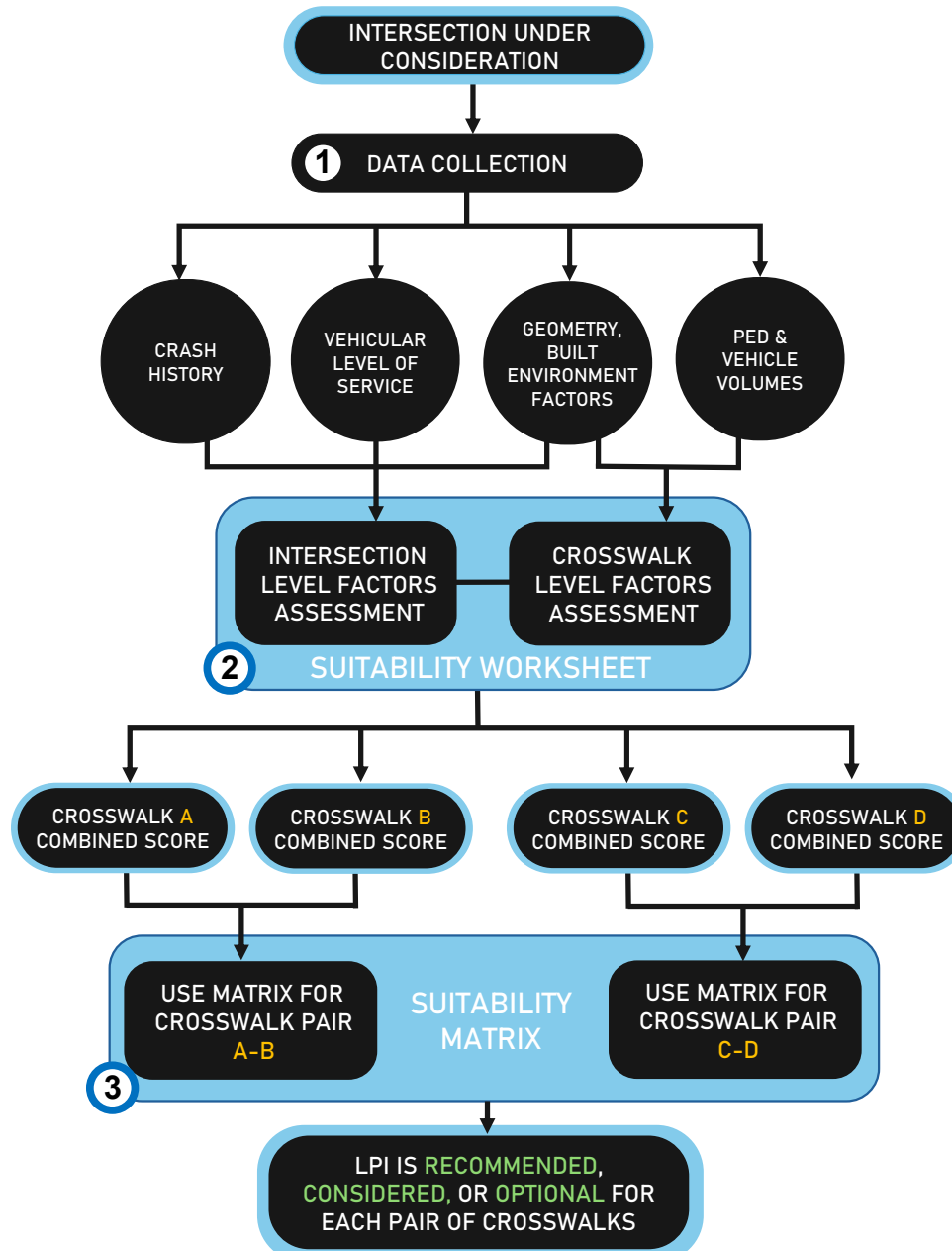


Conflicting Turning Vehicle = Right Turns + Left Turns

Summary of Conflict Analysis

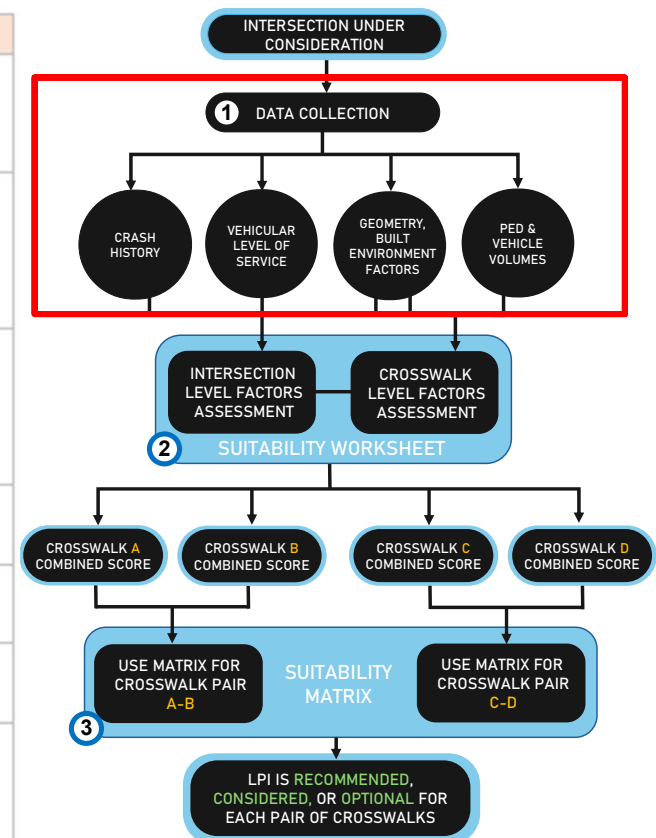
- Analyzed conflicts at 8 Phoenix crosswalks, focusing on $PET \leq 5$ seconds.
- Significant reduction in total conflicts and severity after LPI implementation.
- NB regression models predicted 10-15% reduction in total conflicts and ~45-50% reduction in severe conflicts.
- Ordered logit model also confirmed significant reduction in high and medium severity conflicts.
- Factors like vehicle direction, driver violations, evasive actions and pedestrian group size influenced conflict severity

LPI Guidance Overview



LPI Guidance: Data Collection

Factors	Data Collection Details	Data Collection Methods
Crash History	Number of turning vehicle-pedestrian crashes in the last 5 years in the intersection	Review crash records with appropriate filters.
Pedestrian Volume	Hourly pedestrian volume for each crosswalk	Collect Oct - April, weekday (non-holiday), 10 hrs (7AM-5PM) average preferred. If not in downtown, an average volume from 11AM-2PM is acceptable.
Turning Vehicle Volume	Hourly left and right turning vehicles for each crosswalk	Collect Oct - April, weekday (non-holiday), 10 hrs (7Am-5PM) average preferred. If not in downtown, an average volume from 11AM-2PM is acceptable.
T intersection	Crosswalk parallel with base of T-intersection	Site visit or Google Maps Satellite View
One-Way Street	Presence of one-way street in any approach	Site visit or Google Maps Satellite View
School and Transit Stop Distance	Measure distance from the center of intersection	Measurement using Google Maps
Sight Obstruction	Irregular intersection geometry, obstructions buildings, base of a bridge, trees, blinding sun angle, inferior lighting condition	Site visit or Google Maps Street View
Level of Service (LOS)	Calculate or estimate change in LOS	Traffic analysis software or manual calculations using traffic data

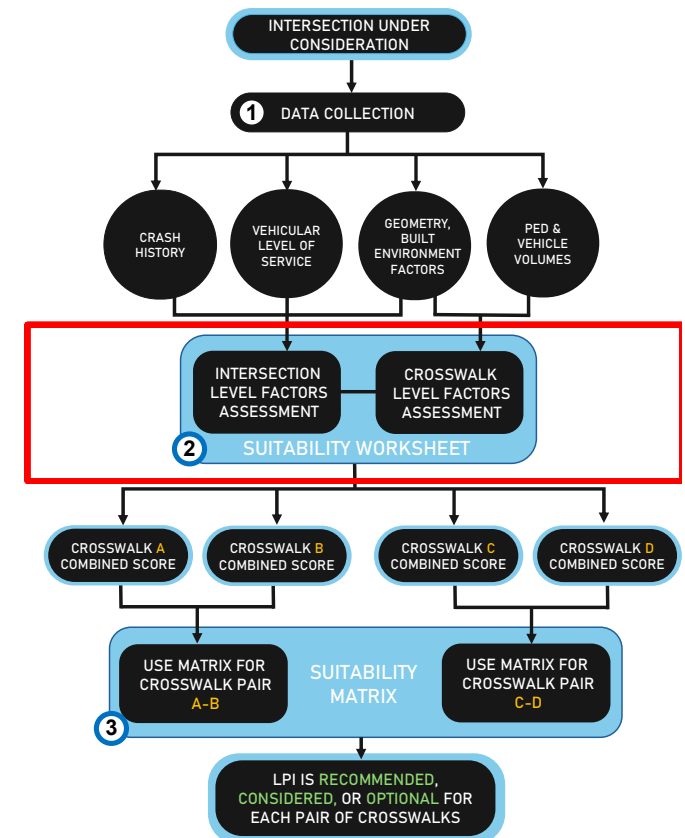


LPI Guidance: Worksheet, Intersection

Intersection Level Assessment			
Factor	Score	Score Allocation	Background
Turning Vehicle - Pedestrian Crashes in recent 5 years	0 to 3	Three crashes or more = 3 Two crashes = 2 One crash = 1 None = 0	Through the literature, an LPI is shown to be effective in reducing this crash type. ¹
One Way Street	0 to 1	One or more approaches = 1 No approach = 0	Vehicles turning left on a one-way street do not need to wait for and yield to vehicles in the opposing direction. ¹
Intersection Distance from School	0 to 2	Less than 2800 ft = 2 More than 2800 ft = 0	Intersections within 2800 ft of a school are found to have increase probability of turning vehicle-pedestrian crashes. ^{2, 3, 4}
Intersection Distance from Transit Stop	0 to 1	Less than 300 ft = 1 More than 300 ft = 0	Intersections within 300 ft of transit stops are found to have increase probability of turning vehicle-pedestrian crashes. ^{2, 3}
Travel Delay	-2 to 0	One approach goes below LOS D = -1 Whole intersection goes below LOS D = -2	High level of travel delay can reduce LPI effectiveness by causing driver frustration and risky behavior. This should be calculated using a preferred analysis software. ¹

Notes:

1. This is used in other published guidance.
2. This is based upon research conducted by NAU.
3. Distance is measured from the center of the intersection, as the crow flies.
4. School is defined as public and K-12 charter schools

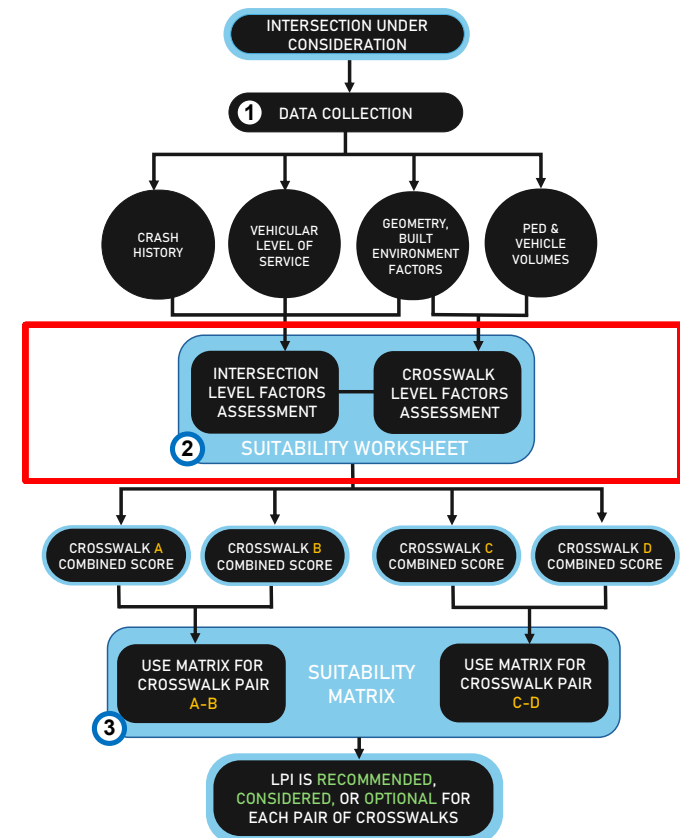


LPI Guidance: Worksheet, Crosswalk

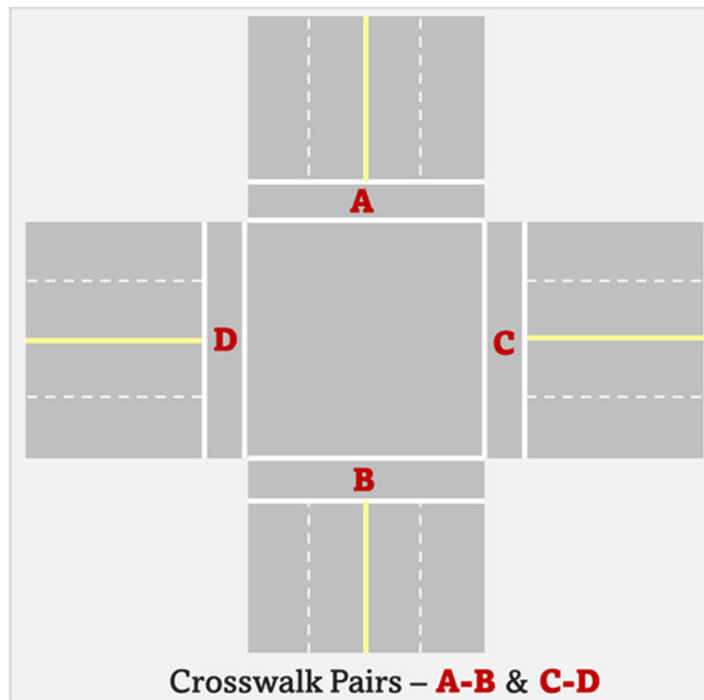
Crosswalk Level Assessment			
Factor	Score	Score Allocation	Reason
Pedestrian Volume (per hour)	0 to 3	More than 100 = 3 51 to 100 = 2 6 to 50 = 1 Less than 6 = 0	Higher pedestrian volumes increase the likelihood of pedestrian-vehicle interactions. ²
Turning Vehicle Volume (vph)	0 to 3	More than 600 = 3 401 to 600 = 2 101 to 400 = 1 Less than 101 = 0	Higher volumes of turning vehicles increase the pedestrian-vehicle interactions. ²
T intersection	0 to 1	Crosswalk parallel with stalk of T intersection = 1 No = 0	In crosswalk parallel with the stalk of a T-intersection, all approaching vehicles turn left or right, with no need to wait for opposing traffic. ¹
Sight Obstruction or Skewed Approach	0 to 1	Yes = 1 No = 0	A sight obstruction or skewed approach can make it more challenging for a vehicle to see a pedestrian on the corner. ¹
Highest Score	15		

Notes:

1. This is used in other published guidance.
2. This is based upon research conducted by NAU.
3. Distance is measured from the center of the intersection, as the crow flies.
4. School is defined as public and K-12 charter schools



LPI Guidance: Suitability Matrix



Crosswalk	A (or C)			
	Score	0-4	5-10	11-15
B (or D)	0-4	Optional	Optional	Considered
	5-10	Optional	Considered	Recommended
	11-15	Considered	Recommended	Recommended

Optional

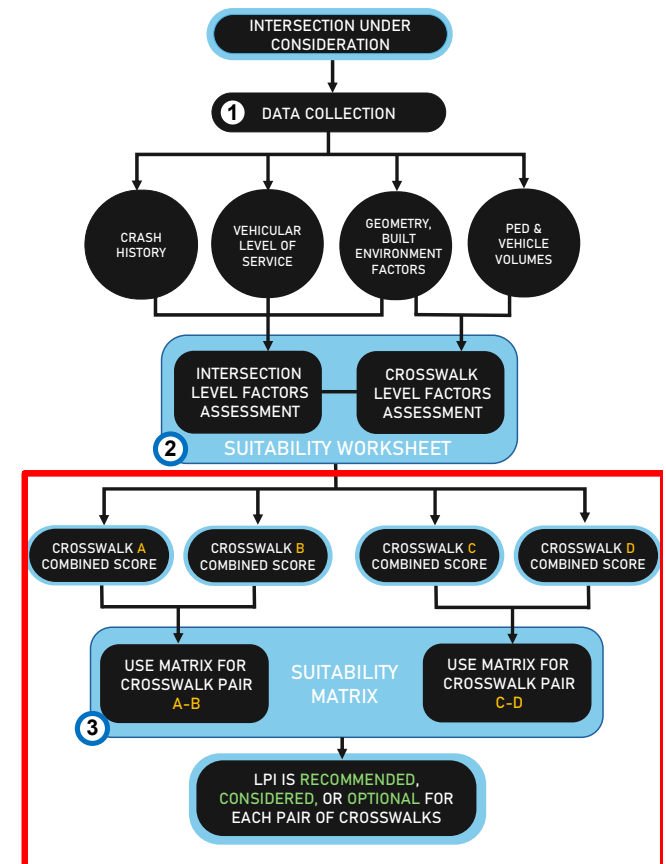
Minor safety benefit expected

Considered

Moderate safety benefit expected

Recommended

High safety benefit expected



LPI Guidance: Other Items

1. If there is no adjacent crosswalk, use 0-4 for Optional, 5-10 for Considered, and 11-15 for Recommended.
2. If not already installed, APS units should be used with all LPI implementations.
3. Due to the need for the voice prompts on APS units to be consistent to cross a given street, an LPI must be installed in pairs such that both crosswalks across a roadway operate with an LPI. Additionally, the 'WALK' indication must start at the same time on both crosswalks.
4. At locations with LPIs, left turn protection should lag the through movement only. Leading left turn protection for a movement across an LPI crosswalk can result in vehicles turning into the LPI.
5. For locations with a Flashing Yellow Arrow, the indication shall remain red until the LPI ends.
6. The guidance provided in this work is based on a 5-second LPI. A shorter or longer LPI duration can be used, based on engineer judgment and MUTCD guidance.
7. On coordinated facilities, the offsets of the adjacent intersections should be adjusted such the vehicle platoons do not arrive during the LPI.
8. Upon implementation of an LPI, signage such as 'Traffic Control Change' or 'New Pedestrian Head Start' may be implemented to inform drivers of the traffic control change.
9. To improve vehicular compliance with the LPI, the following items may be implemented:
 - a. Install visors to shield signal heads perpendicular to the LPI crosswalks so that drivers cannot anticipate the green interval. This is especially useful for skewed locations but can be an option at all locations.
 - b. No Right Turn on Red should be considered on approaches parallel to the LPI crosswalk, as the literature shows it increases the effectiveness of the LPI.
 - c. A 'Turning Vehicles Yield to Pedestrians' sign (R10-15) may be used to increase awareness of pedestrians to drivers.

LPI Guidance: Example

Intersection Level Assessment				
Factor	Score	Score Allocation	North X-Walk	West X-Walk
Turning Vehicle - Pedestrian Crashes in recent 5 years	0 to 3	Three crashes or more = 3 Two crashes = 2 One crash = 1 None = 0	3 pts 3 crashes from 2018-2022	
One Way Street	0 to 1	One or more approaches = 1 No approach = 0	0 pts No one-way streets at location	
Intersection Distance from School	0 to 2	Less than 2800 ft = 2 More than 2800 ft = 0	2 pts School within 2800 ft	
Intersection Distance from Transit Stop	0 to 1	Less than 300 ft = 1 More than 300 ft = 0	1 pt Transit stop within 2800 ft	
Travel Delay	-2 to 0	One approach goes below LOS D = -1 Whole intersection goes below LOS D = -2	0 pts Presumed 0; insufficient data for analysis	

Crosswalk Level Assessment				
Factor	Score	Score Allocation	North X-Walk	West X-Walk
Pedestrian Volume (per hour)	0 to 3	More than 100 = 3 51 to 100 = 2 6 to 50 = 1 Less than 6 = 0	1 pt 28 ped/hr average over 11AM – 2 PM	1 pt 28 ped/hr average over 11AM – 2 PM
Turning Vehicle Volume (vph)	0 to 3	More than 600 = 3 401 to 600 = 2 101 to 400 = 1 Less than 101 = 0	1 pt 343 veh/hr average over 11AM – 2 PM	2 pt 343 veh/hr average over 11AM – 2 PM
T intersection	0 to 1	Crosswalk parallel with stalk of T intersection = 1 No = 0	0 pts Not a T-intersection	0 pts Not a T-intersection
Sight Obstruction or Skewed Approach	0 to 1	Yes = 1 No = 0	0 pts No skew or sight obstruction	0 pts No skew or sight obstruction
Total			8	9



Crosswalk	North (8 pts)			
	Score	0-4	5-10	11-15
South (11 pts)	0-4	Optional	Optional	Considered
	5-10	Optional	Considered	Recommended
	11-15	Considered	Recommended	Recommended

Crosswalk	East (8 pts)			
	Score	0-4	5-10	11-15
West (9 pts)	0-4	Optional	Optional	Considered
	5-10	Optional	Considered	Recommended
	11-15	Considered	Recommended	Recommended

Note: Values for East and South Crosswalks assumed for example purposes

Limitation and Direction for Future Research

- **Crash Analysis**

- Conducted at the intersection level, not individual crosswalk level.
- Absence of observed pedestrian volumes in the crash analysis.

- **Conflict Analysis**

- Manual data reduction may introduce counting or measurement errors.
- Direct sunlight on or near traffic lights caused visibility issues; traffic light status assumed based on surrounding traffic.



Thank You.

Questions/Discussion?

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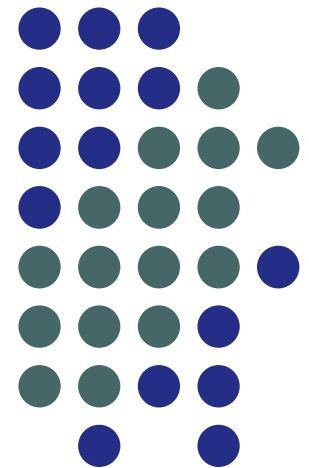
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Conflict Frequency Models

All (3063 Conflicts)					High Severe PET (238 Conflicts)					High Severe PET High Speed (89 Conflicts)				
Coefficients: (Total turning vehicle = Permitted left + Protected Left +					Coefficients:					Coefficients:				
	Estimate	Pr(> z)		LPI Percent Change		Estimate	Pr(> z)		LPI Percent Change		Estimate	Pr(> z)		LPI Percent Change
(Intercept)	1.174	2.00E-16	***		(Intercept)	-1.357	0.0002	***		(Intercept)	-2.341	8.23E-06	***	
LPI_indicator	-0.157	0.00881	**	-14.53%	LPI_indicator	-0.673	8.57E-05	***	-48.97%	LPI_indicator	-0.690	0.00568	**	-49.86%
Total_Turning_Vehicle	0.003	2.00E-16	***		Total_Turning_Vehicle	0.004	6.74E-12	***		Total_Turning_Vehicle	0.004	1.77E-05	***	
Pedestrian_Volume	0.022	2.00E-16	***		Pedestrian_Volume	0.012	0.00931	**		Pedestrian_Volume	0.018	0.00333	**	
Coefficients:					Coefficients:					Coefficients:				
	Estimate	Pr(> z)				Estimate	Pr(> z)				Estimate	Pr(> z)		
(Intercept)	1.030	<2e-16	***		(Intercept)	-1.626	1.19E-05	***		(Intercept)	-2.673	7.89E-07	***	
LPI_indicator	-0.112	0.0474	*	-10.62%	LPI_indicator	-0.585	0.0004	***	-44.29%	LPI_indicator	-0.632	0.00972	**	-46.83%
Permitted_Left_&Right	0.004	<2e-16	***		Permitted_Left_&Right	0.006	1.41E-13	***		Permitted_Left_&Right	0.006	1.69E-06	***	
Pedestrian_Volume	0.022	<2e-16	***		Pedestrian_Volume	0.013	0.0045	**		Pedestrian_Volume	0.020	0.00121	**	
Coefficients:					Coefficients:					Coefficients:				
	Estimate	Pr(> z)				Estimate	Pr(> z)				Estimate	Pr(> z)		
(Intercept)	1.155	<2e-16	***		(Intercept)	-1.129	0.00733	**		(Intercept)	-2.3E+00	0.00025	***	
LPI_indicator	-0.132	0.0201	*	-12.40%	LPI_indicator	-0.690	5.22E-05	***	-49.82%	LPI_indicator	-7.1E-01	0.00541	**	-50.92%
Permitted_Left	0.002	0.0586	.		Permitted_Left	-0.002	0.68658			Permitted_Left	4.1E-05	0.99401		
Total_Right_Turning_Vehicle	0.004	<2e-16	***		Total_Right_Turning_Vehicle	0.006	8.17E-15	***		Total_Right_Turning_Vehicle	5.7E-03	9.38E-07	***	
Pedestrian_Volume	0.021	<2e-16	***		Pedestrian_Volume	0.009	0.04575	*		Pedestrian_Volume	1.7E-02	0.00836	**	

Crash Analysis Results

Variable	Turning Vehicle-Pedestrian Crashes Expected % Change
T intersection Indicator	51.59%
One way street (1 or more approaches)	-71.58%
Intersection skewed $\geq$ ~15 deg.	25.22%
Site Obstruction Present on 1 or More Corners	43.36%
Crosswalk on 1 and 2 legs (ref.)	
Crosswalk on 3 legs	146.47%
Crosswalk on 4 legs	410.83%
Average Crosswalk Length (ft)	0.61%
No Ped Push-buttons Present (ref.)	
Ped Push-button on 1 or More Legs	-22.46%
Exclusive RT Lanes Not Present (ref.)	
Exclusive RT Lanes Present on 1+ Approaches	16.28%
NRTOR Not Present (ref.)	
NRTOR Restriction Present on 1+ Approaches	2.00%
Major Rd AADT AADT 0-15,000 vpd (ref.)	
Major Rd AADT 15,001-30,000 vpd	19.98%
Major Rd AADT 30,001-45,000 vpd	32.62%
Major Rd AADT 45,001+ vpd	29.46%
Minor Rd AADT 0-5,000 (ref.)	
Minor Rd AADT 5,001-15,000 vpd	41.55%
Minor Rd AADT 15,001-30,000 vpd	95.24%
Minor Rd AADT 30,001+ vpd	131.31%
Intersection Distance from School 851m+ (ref.)	
Intersection Distance from School 201-850m	50.63%
Intersection Distance from School $\leq$ 200m	49.22%
Intersection Distance from Transit Stop 301+ ft (ref.)	
Intersection Distance from Transit Stop 0-300 ft	76.38%
Gross Activity Density 0-5 persons+jobs/acre (ref.)	
Gross Activity Density 5.1-15 persons+jobs/acre	189.07%
Gross Activity Density 15.1-30 persons+jobs/acre	346.76%
Gross Activity Density 30+ persons+jobs/acre	205.75%